National Technical University of Athens

Department of Electrical and Computer Engineering

Modeling Smart Electric Vehicle Charging

A Multi-Level Intelligence Approach with Renewable Energy Integration

Diploma Thesis

by

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Περίληψη

Η παρούσα διπλωματική εργασία εξετάζει την έξυπνη φόρτιση ηλεκτρικών οχημάτων σε κτίρια γραφείων με ενσωμάτωση φωτοβολταϊκών, με στόχο να αξιολογήσει πώς διαφορετικά επίπεδα ευφυίας επηρεάζουν το κόστος, τη χρήση της τοπικής ανανεώσιμης ενέργειας και την τήρηση λειτουργικών περιορισμών δικτύου. Η φόρτιση στον χώρο εργασίας αποκτά όλο και μεγαλύτερη σημασία, καθώς η αύξηση των ηλεκτρικών οχημάτων επιβαρύνει τα κτιριακά και δικτυακά συστήματα, ιδιαίτερα όταν η φόρτιση πραγματοποιείται χωρίς κάποιο αυτοματισμό. Τα κτίρια γραφείων αποτελούν κατάλληλο πεδίο εφαρμογής, επειδή συνδυάζουν προβλέψιμα ωράρια εργασίας, μεγάλες περιόδους στάθμευσης και δυνατότητα αξιοποίησης ενέργειας από φωτοβολταϊκά.

Για τον σκοπό αυτό αναπτύχθηκε ένα μοντέλο που περιλαμβάνει το ηλεκτρικό φορτίο του κτιρίου, την παραγωγή από φωτοβολταϊκό σύστημα, αποθήκευση ενέργειας, στόλο ηλεκτρικών οχημάτων και περιορισμούς δικτύου. Η μεθοδολογία βασίζεται σε μελέτη περίπτωσης με ρεαλιστικά δεδομένα κατανάλωσης, ηλιακής ακτινοβολίας και τιμών ηλεκτρικής ενέργειας. Εξετάζονται μια απλή στρατηγική βασισμένη σε κανόνες και μια στρατηγική ενισχυτικής μάθησης με Q-learning, ενώ διερευνάται και η επέκταση σε λειτουργία Vehicle-to-Grid, όπου τα οχήματα μπορούν να επιστρέφουν ενέργεια προς το κτίριο ή το δίκτυο. Η αξιολόγηση πραγματοποιείται με οικονομικά, ενεργειακά, λειτουργικά και υπολογιστικά κριτήρια.

Τα αποτελέσματα δείχνουν ότι η συντονισμένη φόρτιση βελτιώνει τη συνολική λειτουργία του συστήματος σε σχέση με μια απλή ανελαστική στρατηγική. Η προσέγγιση με απλούς κανόνες επιτυγχάνει αξιόπιστη λειτουργία με χαμηλή πολυπλοκότητα και πλήρη συμμόρφωση με τους βασικούς περιορισμούς. Η προσέγγιση με Q-learning προσφέρει επιπλέον βελτίωση στο οικονομικό αποτέλεσμα και καλύτερη διαχείριση της ενέργειας από το δίκτυο, αλλά απαιτεί μεγαλύτερη υπολογιστική προσπάθεια και διαδικασία εκπαίδευσης. Η συμβολή των φωτοβολταϊκών αποδεικνύεται θετική, καθώς μειώνει την εξάρτηση από το δίκτυο και ενισχύει την τοπική αξιοποίηση ανανεώσιμης ενέργειας, αν και το συνολικό όφελος εξαρτάται από το μέγεθος του συστήματος σε σχέση με τη ζήτηση φόρτισης.

Η ανάλυση του Vehicle-to-Grid δείχνει ότι η τεχνική δυνατότητα δεν μεταφράζεται απαραίτητα σε οικονομικό όφελος υπό όλες τις συνθήκες αγοράς. Όταν η διαφορά μεταξύ τιμών αγοράς και πώλησης δεν είναι επαρκής, η αποφόρτιση δεν είναι ελκυστική και τα οφέλη του V2G παραμένουν περιορισμένα. Συνολικά, η εργασία καταλήγει ότι η έξυπνη φόρτιση με ενσωμάτωση φωτοβολταϊκών αποτελεί μια πρακτική και αποτελεσματική λύση για κτίρια γραφείων, ενώ οι πιο προηγμένες στρατηγικές ελέγχου και το V2G αποκτούν αξία κυρίως όταν υποστηρίζονται από κατάλληλες τιμολογιακές και λειτουργικές συνθήκες.

Λέξεις κλειδιά: Ηλεκτρικά οχήματα, έξυπνη φόρτιση, ανανεώσιμες πηγές ενέργειας, βελτιστοποίηση, Vehicle-to-Grid, κτίρια γραφείων

Abstract

This thesis examines the smart charging of electric vehicles (EVs) in office buildings with integrated photovoltaics (PV), aiming to evaluate how different levels of intelligence affect costs, the utilization of local renewable energy, and adherence to grid operational constraints. Workplace charging is becoming increasingly important, as the growth of EVs places a burden on building and grid systems, particularly when charging occurs without automation. Office buildings serve as an ideal field of application because they combine predictable working hours, long parking durations, and the potential to utilize energy from PV systems.

To this end, a model was developed that includes the building's electrical load, PV system generation, energy storage, an EV fleet, and grid constraints. The methodology is based on a case study using realistic consumption, solar radiation, and electricity price data. A simple rule-based strategy and a reinforcement learning strategy using Q-learning are examined, while the expansion into Vehicle-to-Grid (V2G) operation—where vehicles can return energy to the building or the grid—is also explored. The evaluation is conducted based on economic, energetic, operational, and computational criteria.

The results indicate that coordinated charging improves overall system performance compared to a simple inelastic strategy. The rule-based approach achieves reliable operation with low complexity and full compliance with basic constraints. The Q-learning approach offers further improvement in economic outcomes and better management of grid energy but requires greater computational effort and a training process. The contribution of photovoltaics proves positive, as it reduces grid dependence and enhances local renewable energy utilization, although the overall benefit depends on the system's size relative to charging demand.

The analysis of Vehicle-to-Grid shows that technical capability does not necessarily translate into economic benefit under all market conditions. When the price spread between buying and selling is insufficient, discharging is unattractive, and V2G benefits remain limited. Overall, the thesis concludes that smart charging with PV integration is a practical and effective solution for office buildings, while more advanced control strategies and V2G gain value primarily when supported by appropriate pricing and operational conditions.

Keywords: Electric vehicles, smart charging, renewable energy, optimization, Vehicle-to-Grid, office buildings

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Chapter 1

Introduction

1.1 Background and Motivation

Recent climate regulations have led to the rise of the adoption of electric vehicles. More and more people are opting for an electric vehicle for their daily commute. According to studies most of the time spend in commute is on the way to and from the office. Also most people spend a third of their time in offices and with them their cars, which can be charged during the work day. The car batteries can be also used in time of peak consumption to give electricity back to the building or to the grid. Since the charging will happne throughout the day, it is logical to create smart charging algorithms that can utilise information from solar power generation, pricing spreads around the day and grid capacity limits in order to optimise the charging shedule. Office building offer also a huge rooftop space for the installation of solar panels, energy produced by these panels can be used both for the building consumption as well as for the charging of the electric vehicles. The cost of electricity produced is zero and this helps reduce the overall cost. As such the utilization of renewable energy is the most important aspect of the charging algorithm.

1.2 Literature Review

Currently there are many different algorithms that are used for the smart scheduling of the EV fleet. Early studies such as Richardson [1] used linear programming in order to decide which vehicle will be charged and when in 15 min intervals, taking into account network limitations. Further studies included the use of reinforcement learning to increase grid stability, optimize for pricing and renewables utilization, several studies are included in this paper [2]

1.3 Smart Charging Levels

The algorithms developed in this thesis are separated by their inteligence levels. Level 0 is a simple rule based algorithm that just prioritises solar renewable energy, respecting grid constraints and also target charging SOC of the EV battery, it charges without respecting future price levels. Level 1 is a Q-learning algorithm that uses a reward function to include information about the price levels and also the availability of solar energy. Level 2 is a Q-learning algorithm that includes also the action of discharging energy back to the grid.The rest chapters focus on different parts of the algorithms.

1.4 Thesis Structure

Chapter 2 focus more on the model that describe the office the evs the grid and the solar panels. Chapter 3 focus on the data for the office building consumption, electricity pricing and solar generation. Chapter 4 focus on the simulation using each algorithm, collecting and comparing the results. Chapter 5 focus on the conclusion of the results and the recommendations for the future work.

Chapter 2

Methodology

This chapter describes the methodology used in order to simulate the real environment. It includes the models as well as the equations used to describe the different components. The differnt algorithms are also described in this chapter in order of inteligence levels.

2.1 System Components

2.1.1 Office Building Model

The office building is modeled first by it's energy consumption. The stationary battery of the building is also represented by a battery model. The solar panels are also represented by a solar panel model. The grid is also represented by a grid model.

Energy Consumption

For the building energy consumption $C_b(t)$, we utilize the ELMAS dataset (European Load Model for Aggregated Sectors), specifically the Office cluster (#5). Unfortunatelly the column has only mean data for the classes of the cluster, so we calculated a sample building annual energy. Then we extracted this data in the `Time_series_18_clusters.csv` file. THe following processing steps where followed.

Let L_t be the mean hourly energy load of the different classes of the Office cluster (T = 8,736 hours for year 2018) with yearly energy:

$$E_{cluster} = \sum_{t=1}^{T} L_t \tag{2.1}$$

To be able to represent a single office building, we calculate the per-customer annual energy for each class c belonging to the Office cluster column:

$$e_c = \frac{E_c}{N_c} \tag{2.2}$$

where E_c is the annual energy and N_c the number of customers in class c. The 75th percentile is used to scale the energy to represent a medium to big office.

$$E_{target} = Q_{0.75}(\{e_c\}) = 58{,}831 \text{ kWh} \tag{2.3}$$

The energy profile L_t is then scaled by using the energy target of the single building.

$$L_t^{scaled} = L_t \cdot \frac{E_{target}}{E_{cluster}} \tag{2.4}$$

To be able to simulate a more realistic schenario we introduce also noise to the energy profile. We use a daily factor α_d (constant within each day) and an hourly factor β_t:

$$\alpha_d \sim \mathcal{N}(1, \sigma_d^2), \quad \sigma_d = 0.08, \quad \alpha_d \in [0.7, 1.3] \tag{2.5}$$

$$\beta_t \sim \mathcal{N}(1, \sigma_h^2), \quad \sigma_h = 0.03, \quad \beta_t \in [0.9, 1.1] \tag{2.6}$$

The synthetic series before re-normalization:

$$L_t^{syn} = L_t^{scaled} \cdot \alpha_d(t) \cdot \beta_t \tag{2.7}$$

Finally, as we did in the previous step, we re-scale the series to preserve the annual energy.

$$C_b(t) = L_t^{syn} \cdot \frac{E_{target}}{\sum_{t=1}^{T} L_t^{syn}} \tag{2.8}$$

The single-office profile has a mean load of 6.73 kW and a peak load of 11.92 kW.

Solar Photovoltaic System

Solar PV system is modeled by the following equation:

$$P_{solar}(t) = A_{panel} \cdot \eta_{panel} \cdot I(t) \tag{2.9}$$

where:

- A_{panel} = solar panel area (m^2)
- η_{panel} = panel efficiency
- $I(t)$ = solar irradiance at hour t (W/m^2)

Building Battery Storage

The building has a battery is modeled as follows.

$$P_{dis,t} = \min(P_{max},\ (\mathrm{SoC}_{t-1} - \mathrm{SoC}_{min}) \cdot \eta) \tag{2.10}$$

$$P_{ch,t} = \min(P_{max},\ C - \mathrm{SoC}_{t-1}) \tag{2.11}$$

$$\mathrm{SoC}_t = \mathrm{SoC}_{t-1} + \min(P_{ch,t},\ E_{excess}(t)) \tag{2.12}$$

$$\mathrm{SoC}_t = \mathrm{SoC}_{t-1} - \min\left(\frac{P_{dis,t}}{\eta},\ \frac{E_{missing}(t)}{\eta}\right) \tag{2.13}$$

where:

- P_{max} = maximum charge/discharge power [kW]
- C = battery capacity [kWh]
- SoC_{min} = minimum allowable state of charge
- η = round-trip efficiency
- $E_{excess}(t) = \max(0,\ P_{solar}(t) - C_b(t))$ = surplus solar energy
- $E_{missing}(t) = \max(0,\ C_b(t) - P_{solar}(t))$ = energy deficit

Constraints:

$$\mathrm{SoC}_{min} \leq \mathrm{SoC}_t \leq C \tag{2.14}$$

$$0 \leq P_{ch,t} \leq P_{max} \tag{2.15}$$

$$0 \leq P_{dis,t} \leq P_{max} \tag{2.16}$$

Net Energy Demand

The building's net energy demand (positive = needs grid, negative = surplus):

$$E_{net}^{building}(t) = C_b(t) - P_{solar}(t) - E_{discharge}^{b}(t) + E_{charge}^{b}(t) \tag{2.17}$$

2.1.2 Electric Vehicle Fleet Model

Individual EV Characteristics

Each EV i in the fleet has the following info:

- Battery capacity: B^i_{cap} [kWh]
- State of charge: $\text{SoC}^i(t)$ [dimensionless, 0-1]
- Maximum charge rate: $P^i_{max,charge}$ [kW]
- Maximum discharge rate: $P^i_{max,discharge}$ [kW] (for V2G)
- Arrival time: $t^i_{arrival}$ [hour]
- Departure time: t^i_{target} [hour]
- Desired SoC: $\text{SoC}^i_{desired}$ [dimensionless]

EV Battery Dynamics

The state of charge evolves for EV i following the equation:

$$\text{SoC}^i(t+1) = \text{SoC}^i(t) + \frac{\eta_{charge} \cdot E^i_{charge}(t) - E^i_{discharge}(t)/\eta_{discharge}}{B^i_{cap}} \tag{2.18}$$

where η_{charge} and $\eta_{discharge}$ are charging and discharging efficiencies (typically 0.95).

Constraints

Each EV must satisfy:

$$\text{SoC}_{min} \leq \text{SoC}^i(t) \leq 1.0 \quad \forall t \tag{2.19}$$

$$\text{SoC}^i(t^i_{target}) \geq \text{SoC}^i_{desired} \tag{2.20}$$

$$E^i_{charge}(t) = 0 \quad \text{if } t < t^i_{arrival} \text{ or } t \geq t^i_{target} \tag{2.21}$$

Constraint (2.20) makes sure that each EV reaches its desired charge level before departure.

2.1.3 Grid Model

Dynamic Pricing

For the simulations we use dynamic pricing with different price each hour

$$\pi_{buy}(t) = \text{price\_profile}[t \bmod 24] \quad [\text{€/kWh}] \tag{2.22}$$

For V2G scenarios, the selling prices would be 80% of the buying prices

$$\pi_{sell}(t) = \text{sell\_price\_profile}[t \bmod 24] \quad [€/\text{kWh}] \tag{2.23}$$

Typically, $\pi_{sell}(t) \leq \pi_{buy}(t)$ (80-90% of purchase price).

Grid Capacity Constraint

Grid provides a certain capacity that can be used by the building for charging

$$E_{grid}^{building}(t) + \sum_{i=1}^{N_{EV}} E_{grid}^{i}(t) \leq P_{grid,max}(t) \tag{2.24}$$

where $P_{grid,max}(t)$ represents the hourly grid capacity limit.

2.2 Optimization Problem Formulation

2.2.1 Objective Function

The goal is to minimize the total electricity cost:

$$\min \sum_{t=0}^{T-1} \left[\pi_{buy}(t) \cdot E_{grid}^{total}(t) - \pi_{sell}(t) \cdot E_{V2G}^{total}(t)\right] \tag{2.25}$$

where:

- $E_{grid}^{total}(t)$ = total energy purchased from grid at hour t
- $E_{V2G}^{total}(t)$ = total energy sold back to grid (V2G only)
- T = simulation duration (24 hours)

2.2.2 Energy Balance

For each hour t, the energy equation must be satisfied:

$$C_b(t) + \sum_{i=1}^{N_{EV}} E_{charge}^{i}(t) = P_{solar}(t) + E_{grid}^{total}(t) + \sum_{i=1}^{N_{EV}} E_{discharge}^{i}(t) \tag{2.26}$$

2.3 Level 0: Baseline (Uncontrolled Charging)

2.3.1 Algorithm Description

The simple algorithm follows priorities first the remaining solar energy and respects grid capacity, each car charges at max rate upon arrival until its desired state of charge is reached:

Algorithm 1 Simple Rule-Based Charging

for each hour t **do**
    Calculate building surplus: $E_{surplus}(t) = \max(0, P_{solar}(t) - C_b(t))$
    Calculate available grid capacity: $P_{available}(t) = P_{grid,max}(t) - \max(0, C_b(t) - P_{solar}(t))$
    $E_{remaining} \leftarrow E_{surplus}(t)$
    for each EV i (in order) **do**
        if $t^i_{arrival} \leq t < t^i_{target}$ and $\text{SoC}^i(t) < \text{SoC}^i_{desired}$ **then**
            $E^i_{needed} = \min(P^i_{max,charge}, (\text{SoC}^i_{desired} - \text{SoC}^i(t)) \cdot B^i_{cap})$
            $E_{from\_surplus} = \min(E^i_{needed}, E_{remaining})$
            $E_{from\_grid} = \min(E^i_{needed} - E_{from\_surplus}, P_{available}(t))$
            $E_{total} = E_{from\_surplus} + E_{from\_grid}$
            Charge EV i with E_{total}
            Update $E_{remaining}$ and $P_{available}(t)$
        end if
    end for
end for

2.3.2 Mathematical Formulation

Energy for EV i at hour t:

$$E^i_{charge}(t) = \min\left(P^i_{max,charge},\ (\text{SoC}^i_{desired} - \text{SoC}^i(t)) \cdot B^i_{cap},\ E_{available}(t)\right) \tag{2.27}$$

where $E_{available}(t)$ contains both building extra energy and remaining grid capacity.

2.4 Level 2: Intelligent Optimization Algorithms

2.4.1 Reinforcement Learning (Q-Learning)

State Space

The state for EV i at hour t is represented by the following:

$$s^i(t) = (\text{EV\_id},\ \text{SoC}^i_{bin}(t),\ t,\ \text{grid\_availability}) \tag{2.28}$$

where $\text{SoC}^i_{bin}(t)$ is the discretized SoC (bins of 0.1).

Action Space

For charge-only systems:

$$\mathcal{A} = \{\text{charge}, \text{standby}\} \tag{2.29}$$

For V2G-enabled systems:

$$\mathcal{A}_{V2G} = \{\text{charge}, \text{discharge}, \text{standby}\} \tag{2.30}$$

Reward Function

The reward function is the following:

$$\begin{aligned} R(s,a,s') = & -\alpha_1 \cdot E^i_{grid}(t) \cdot \pi_{buy}(t) \quad \text{(minimize grid cost)} \\ & + \alpha_2 \cdot E^i_{solar}(t) \quad \text{(maximize solar usage)} \\ & + \alpha_3 \cdot \nVdash[\text{SoC}^i(t^i_{target}) \geq \text{SoC}^i_{desired}] \quad \text{(target achievement)} \\ & - \alpha_4 \cdot \max(0, \text{SoC}^i_{desired} - \text{SoC}^i(t^i_{target})) \quad \text{(penalty for shortfall)} \\ & - \alpha_5 \cdot \nVdash[\text{SoC}^i(t) < \text{SoC}_{min}] \quad \text{(constraint violation)} \end{aligned} \tag{2.31}$$

where $\alpha_1, \ldots, \alpha_5$ are weights.

Q-Learning Update Rule

The Q-value is updated using the temporal difference method:

$$Q(s,a) \leftarrow Q(s,a) + \alpha \left[R(s,a,s') + \gamma \max_{a'} Q(s',a') - Q(s,a) \right] \tag{2.32}$$

where:

- α = learning rate (0.1-0.15)
- γ = discount factor (0.95-1.0)
- s' = next state after taking action a in state s

Policy Execution

After training, the policy selects actions greedily:

$$a^*(s) = \arg\max_{a \in \mathcal{A}} Q(s,a) \tag{2.33}$$

2.5 Level 2: Vehicle-to-Grid (V2G)

2.5.1 Bidirectional Energy Flow

V2G expands the optimization to include discharge decisions:

$$E^i_{net}(t) = E^i_{charge}(t) - E^i_{discharge}(t) \tag{2.34}$$

2.5.2 V2G Constraints

Additional constraints for V2G are the following:

$$E^i_{discharge}(t) \leq (\mathrm{SoC}^i(t) - \mathrm{SoC}^i_{desired}) \cdot B^i_{cap} \quad \text{(preserve target SoC)} \tag{2.35}$$

$$E^i_{charge}(t) \cdot E^i_{discharge}(t) = 0 \quad \text{(no simultaneous charge/discharge)} \tag{2.36}$$

2.5.3 Revenue Calculation

Total benefit with V2G:

$$\mathrm{Benefit}_{V2G} = -\sum_{t=0}^{T-1} \left[\pi_{buy}(t) \cdot E^{total}_{grid}(t) - \pi_{sell}(t) \cdot \sum_{i=1}^{N_{EV}} E^i_{discharge}(t) \right] \tag{2.37}$$

Positive benefit indicates revenue or cost saving.

2.5.4 V2G Integration with Algorithms

The Q-Learning algorithm is extended to handle V2G by adding the action of discharge and also by adding sell reward to the reward function, also discharge constraints are added.

2.6 Key Assumptions and Limitations

2.6.1 Assumptions

Chapter 3

Case Study: Office Building Scenario

This chapter lays out the office building case study used to test the charging strategies. It also explains where the data comes from and why the key parameters are set the way they are.

The case study models a single office building with rooftop PV, workplace charging for 15 vehicles, a grid connection with capacity limits, and a one-day simulation window.

3.1 Building Energy Profile

3.1.1 Energy Consumption Data

Data Source

The building energy consumption profile is derived from the ELMAS (European Load Model for Aggregated Sectors) dataset, specifically the Office cluster (#5). The dataset file is `office_single_building_timeseries.csv` and the source is the ELMAS file `Time_series_18_clusters.csv` for Office cluster #5. The building type is office/commercial, the target annual energy is 58,831 kWh (the 75th percentile of the per-customer distribution), and the temporal resolution is hourly with 8,760 data points for year 2018.

The profile was built by scaling the cluster load shape to the target annual energy, then added daily and hourly variability. The series is re-normalized so the annual total stays fixed (see Section 2.1.1 for full methodology).

Profile Characteristics

Figure 3.1 shows the hourly consumption pattern:

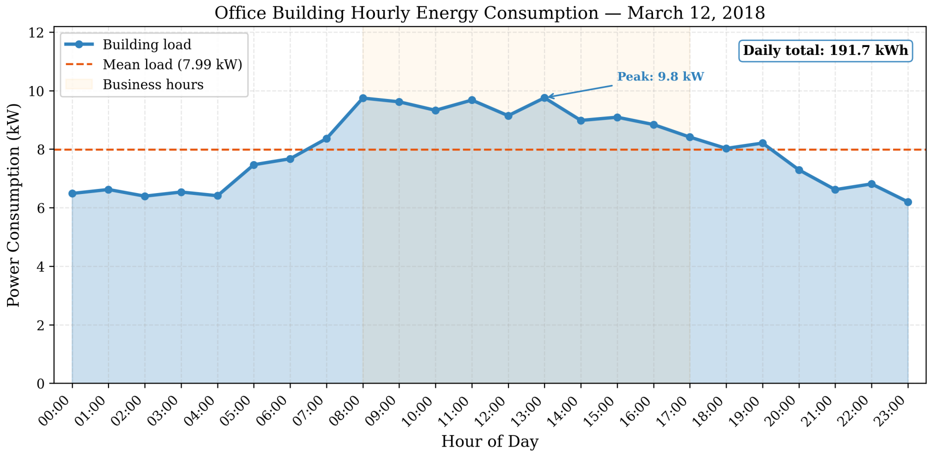

Figure 3.1: Office building hourly energy consumption profile

Annual statistics are a mean load of 6.73 kW, a peak load of 11.92 kW during business hours, an annual energy of 58,831 kWh, and a daily consumption of about 161 kWh (annual divided by 365).

Validation

The consumption profile comes from the ELMAS source data. The per-customer annual energy distribution, with mean 10,429 kWh and P75 = 58,831 kWh, reflects small to medium-sized office buildings. The 75th percentile is used to represent a medium-to-large office. The cluster shape has a clear working-hours peak and a low night baseline, which matches an office profile, and the added noise keeps the curve from looking overly smooth while preserving the annual total.

3.1.2 Solar PV System

System Specifications

The rooftop solar PV system parameters:

Table 3.1: Solar PV system specifications

Parameter	Value
Panel area	150 m^2
Panel efficiency	20%
Peak power	30 kW
Technology	Monocrystalline PERC (JA Solar DeepBlue 3.0 Pro)
Reference module	JAM54S31-405/MR (405 Wp, 20.7% efficiency)
Number of modules	74

The reference module is the JA Solar DeepBlue 3.0 Pro (JAM54S31-405/MR) [3], a widely deployed monocrystalline PERC half-cut cell panel rated at 405 Wp with a module efficiency of 20.7% (STC). The installation comprises 74 units, yielding a combined unit area of approximately 144.5 m^2 ($\approx$ 150 m^2) and a peak system power of 29.97 kW ($\approx$ 30 kW).

Irradiance Data

Solar irradiance data is obtained from PVGIS (Photovoltaic Geographical Information System) [4]. The source is the European Commission Joint Research Centre PVGIS database, using the Paris, France location (48.857°N, 2.352°E, elevation 32 m) and the PVGIS-SARAH3 radiation database. The panel orientation is south-facing (azimuth 0°) with a tilt of 37° (site-optimum). The date is March 12, 2018 as a representative spring day. The temporal resolution is hourly and the data file is `irradiance_pvgis.csv`.

Solar Production Profile

The hourly solar power production:

$$P_{solar}(t) = A_{panel} \cdot \eta_{panel} \cdot I(t) = 150\,\mathrm{m}^2 \cdot 0.20 \cdot I(t) \tag{3.1}$$

Figure 3.2 illustrates the production profile:

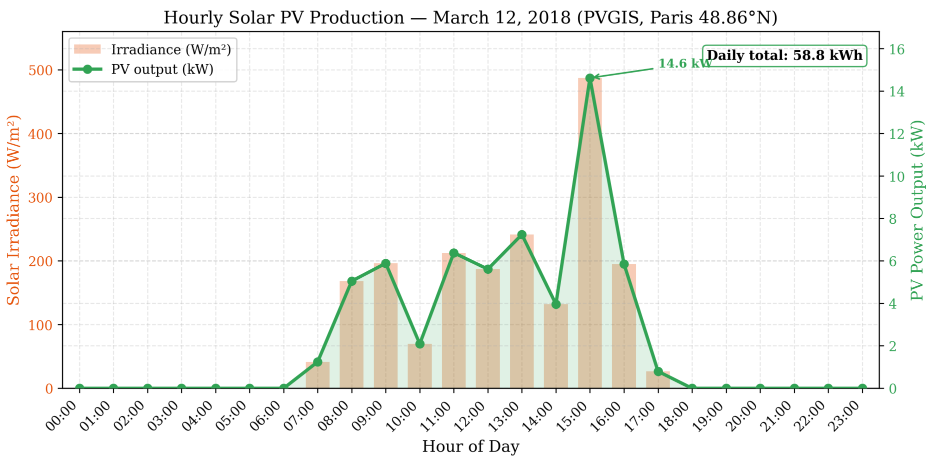

Figure 3.2: Hourly solar PV production profile (March 12, 2018)

Peak production is about 14.6 kW at 15:00, with a peak irradiance of 487 W/m^2. Total daily production is 58.8 kWh. Solar production partially offsets building consumption during business hours (7:00–17:00), with a brief surplus at 15:00. The profile aligns with EV parking hours (6 AM–6 PM) and shows variable irradiance due to sporadic cloud cover typical of Paris in spring.

Selection Justification

March 12, 2018 was selected because it is a spring day with mid-range PV output and variable cloud cover in the Paris PVGIS record. It avoids extreme summer or winter days, and the SARAH3 data for that date are well validated.

3.1.3 Building Battery Storage

The building includes a stationary battery storage system:

Table 3.2: Building battery specifications

Parameter	**Value**
Capacity	80 kWh
Efficiency	95%
Initial SoC	50%
Depth of discharge limit	80%
Max charge rate	50 kW
Max discharge rate	50 kW
Chemistry	Lithium Iron Phosphate (LFP)
Reference product	BYD Battery-Box Premium HVM

The battery parameters are representative of a modular Lithium Iron Phosphate (LFP) system based on the BYD Battery-Box Premium HVM series [5]. The HVM system uses cobalt-free LFP cells with a certified round-trip efficiency of $\geq 96\%$ (HTW Berlin) and offers modular scalability in 2.76 kWh increments. In the simulated configuration, the equivalent of four HVM 19.3 towers ($4 \times 19.32 = 77.3$ kWh ≈ 80 kWh) provides a combined maximum charge/discharge power of approximately 50 kW.

3.2 Electric Vehicle Fleet

3.2.1 Fleet Composition

The simulation supports a fleet of 15 electric vehicles, representing a typical office building with 50-100 employees (assuming 15-30% EV adoption).

Vehicle Specifications

Each EV is modeled with identical technical characteristics:

Table 3.3: Electric vehicle specifications

Parameter	**Value**
Battery capacity	100 kWh
Reference vehicle	Tesla Model 3 Long Range
Max charge rate	22 kW (Level 2 AC charger)
Max discharge rate	22 kW (V2G capable)
Energy consumption	0.18 kWh/km
Depth of discharge limit	90%
Minimum SoC	20%

Justification

The 100 kWh capacity is representative of modern long-range EVs such as the Tesla Model 3 and Hyundai Ioniq 5. A 22 kW charge rate reflects standard Level 2 AC charging that is common in workplace installations. An energy use of 0.18 kWh/km is typical for mid-size EVs, and a 20% minimum SoC prevents deep discharge and limits battery degradation.

3.2.2 Mobility Patterns

Arrival and Departure Times

The EV fleet follows typical office worker patterns:

Table 3.4: EV mobility parameters

Parameter	**Value**
Arrival window	6:00 - 9:00 AM
Departure window	4:00 - 8:00 PM
Parking duration	7-14 hours
Initial SoC range	20-30%
Desired SoC range	70-80%

Each EV's specific arrival and departure times are randomly sampled from the specified windows to simulate realistic variability.

State of Charge Requirements

The initial SoC of 20–30% represents battery level after a morning commute of about 30–50 km. The desired SoC of 70–80% ensures enough charge for the evening commute plus a reserve of roughly 100–150 km.

Justification

The mobility pattern is tied to office schedules. The morning arrival spread from 6–9 AM covers flexible start times, and the evening departure spread from 4–8 PM reflects different end-of-day routines. Long parking durations make it possible to move charging around, while the SoC targets still cover the commute.

3.3 Grid Parameters

3.3.1 Electricity Pricing

Day-Ahead Market Prices

The simulation uses real wholesale day-ahead electricity prices from the EPEX SPOT exchange for the French bidding zone on March 12, 2018. The data was gathered from the ENTSO-E Transparency Platform [6] and the Fraunhofer ISE Energy-Charts API [7].

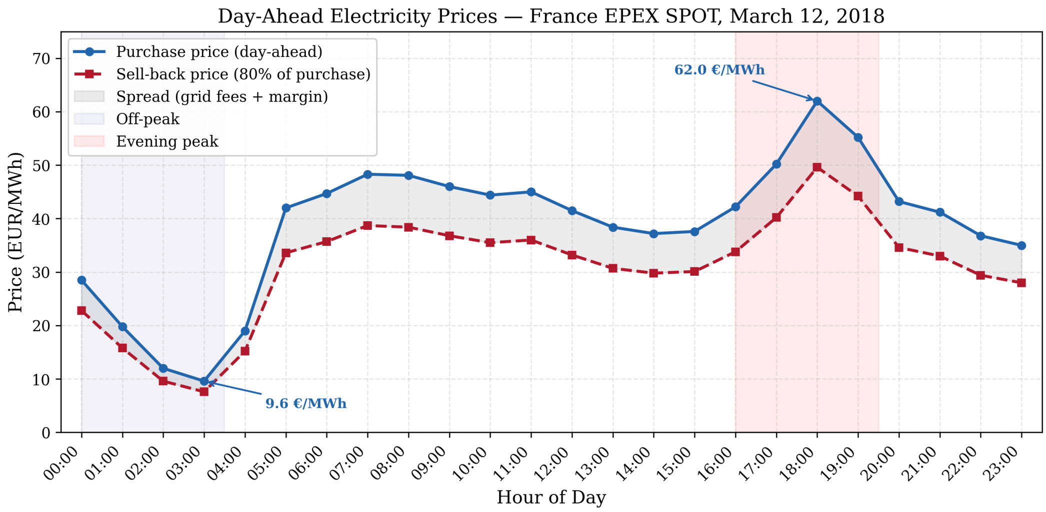

Figure 3.3: Hourly electricity purchase and sell-back prices (France EPEX SPOT, March 12, 2018)

Table 3.5 summarizes the hourly day-ahead prices by time period. The complete hourly data is provided in Appendix B.2.

Table 3.5: Day-ahead electricity prices by time period (France, March 12, 2018)

Time Period	**Purchase (EUR/MWh)**	**Purchase (€/kWh)**	**Sell-back (€/**
Night (00:00–03:00)	9.56–28.46	0.010–0.028	0.008–0.02
Morning ramp (04:00–07:00)	19.04–48.34	0.019–0.048	0.015–0.03
Business hours (08:00–11:00)	44.41–48.05	0.044–0.048	0.036–0.03
Midday (12:00–15:00)	37.24–41.50	0.037–0.042	0.030–0.03
Evening peak (16:00–19:00)	42.23–62.00	0.042–0.062	0.034–0.05
Night (20:00–23:00)	35.04–43.23	0.035–0.043	0.028–0.03
Daily average	**38.67**	**0.039**	**0.031**

Price Profile Characteristics

The day-ahead price curve for March 12, 2018 has the familiar daily shape. There is an off-peak trough between 02:00 and 04:00, with prices of 9.56–12.00 EUR/MWh due to low demand and high nuclear baseload. A morning ramp follows between 05:00 and 08:00, when prices rise from 19.04 to 48.34 EUR/MWh as commercial and industrial loads come online. A daytime plateau from 08:00 to 15:00 keeps prices around 37–48 EUR/MWh, partly due to solar PV and hydro offsetting demand. The evening peak reaches the daily maximum of 62.00 EUR/MWh at hour 19 (6–7 PM) as solar production decreases and residential heating and lighting demand rises. After 20:00, prices gradually fall from 43.23 to 35.04 EUR/MWh as demand subsides and the nuclear fleet provides ample overnight capacity.

Sell-Back Prices (V2G)

For the V2G scenario, sell-back prices are set at 80% of the day-ahead purchase price to reflect grid access and balancing fees, aggregator margin, and round-trip efficiency losses.

Data Source

The electricity price data was obtained from the ENTSO-E Transparency Platform [6] and cross-validated against the Fraunhofer ISE Energy-Charts API [7]. The original market data is from EPEX SPOT SE, the power exchange operating the French day-ahead auction as part of the Single Day-Ahead Coupling (SDAC) mechanism [8].

3.3.2 Grid Capacity Constraints

Capacity Limits

The grid connection has hourly capacity limits:

$$P_{grid,max}(t) = 80\,\text{kW} \quad \forall t \tag{3.2}$$

Justification

The 80 kW limit represents a typical distribution transformer capacity for small commercial buildings. The building baseline consumption is about 8 kW, leaving roughly 70 kW available for EV charging, which is enough for 3–4 simultaneous Level 2 chargers.

Without scheduling, 15 EVs at 22 kW each would require 330 kW, far above the 80 kW limit. Charging therefore has to be staggered across the day.

3.4 Simulation Configuration

3.4.1 Time Parameters

Table 3.6: Simulation time parameters

Parameter	Value
Simulation duration	24 hours
Time step	1 hour
Simulation date	March 12, 2018
Start time	00:00 (midnight)
End time	23:00 (11 PM)

3.4.2 Algorithm Parameters

The reinforcement learning setup uses the following hyperparameters:

Reinforcement Learning

Table 3.7: Q-Learning hyperparameters

Parameter	Charge-Only	V2G
Training episodes	8,000	15,000
Learning rate (α)	0.15	0.15
Discount factor (γ)	0.99	1.0
Exploration rate (ϵ)	0.35	0.10
SoC discretization	0.1 (10% bins)	0.1 (10% bins)

Justification: V2G uses more training episodes because the action space is larger (three actions instead of two). The discount factor is set to $\gamma = 1.0$ in V2G so later discharge opportunities are valued the same as near-term charging. The exploration rate ϵ is lower to emphasize exploitation once the policy stabilizes.

3.5 Data Sources Summary

Table 3.8: Summary of data sources

Data Type	Source	Reference
Building consumption	ELMAS dataset (Office cluster #5)	[9]
Solar irradiance	PVGIS	[4]
Electricity prices	EPEX SPOT / ENTSO-E (France)	[6], [8]
Solar PV modules	JA Solar DeepBlue 3.0 Pro	[3]
Building battery	BYD Battery-Box Premium HVM	[5]
EV specifications	Tesla Model 3	[10]
Charging standards	IEC 61851	[11]

3.6 Simulation Scenarios

3.6.1 Scenario 1: Charge-Only (No V2G)

Configuration: `multi_ev_config.yml`

This scenario evaluates unidirectional charging with 15 EVs that have charge-only capability, the Simple and RL (Q-Learning) algorithms, an 80 kW grid capacity limit, and no discharge allowed.

Purpose: Establish baseline performance for intelligent charging without V2G.

3.6.2 Scenario 2: V2G-Enabled

Configuration: `multi_ev_v2g_config.yml`

This scenario adds bidirectional capability with the same 15 EVs, the Simple and RL algorithms (with a discharge action), sell-back prices set at 80–90% of purchase prices, and discharge constraints that preserve the desired SoC.

Purpose: Quantify additional benefits of V2G technology.

3.7 Evaluation Metrics

The following metrics are used to evaluate each approach.

3.7.1 Economic Metrics

Economic metrics track total daily cost (electricity purchases minus V2G revenue), cost per kWh, and savings versus the Level 0 baseline.

3.7.2 Energy Metrics

Energy metrics capture total grid consumption, renewable utilization (share of charging supplied by solar), and V2G energy discharged to the grid.

3.7.3 Performance Metrics

Performance metrics cover target achievement, average final SoC at departure, grid violations (hours over the capacity limit), and SoC violations (instances below the minimum).

3.7.4 Computational Metrics

Computational metrics report execution time per simulation, training time for learning-based methods, and how performance scales with fleet size.

3.8 Implementation

3.8.1 Software Environment

The simulation is implemented in Python 3.x and uses NumPy for numerical work, Matplotlib for plots, and PyYAML for configuration files.

Table 3.9: Software dependencies

Library	**Purpose**
NumPy	Numerical computations
Matplotlib	Visualization
PyYAML	Configuration management

3.8.2 Code Structure

The code is organized around core models for the building, electric vehicles, and the grid. Charge-only logic lives in `MultiEVChargingSystem`, while V2G logic is handled by `MultiEVV2GChargingSystem`. The `multi_ev_simulator` and `multi_ev_v2g_simulator` scripts run the charge-only and V2G scenarios.

Chapter 4

Results and Analysis

This chapter shows the simulation results for both charge-only and V2G-enabled scenarios. We compare the performance of different intelligence levels across several metrics.

4.1 Overview of Results

Simulations covered two main scenarios. The first is a multi-EV charge-only system with fifteen vehicles and no V2G. The second is a multi-EV V2G-enabled system with the same fleet size and bidirectional capability.

Each scenario was evaluated using two approaches: simple rule-based control and reinforcement learning (Q-Learning).

4.2 Scenario 1: Charge-Only Results

4.2.1 Economic Performance

Total Daily Costs

Table 4.1 summarizes the economic performance of each method:

Table 4.1: Economic comparison for charge-only scenario

Method	Total Cost (€)	Savings vs Simple (%)	Cost/kWh (€)	Grid Energy
Simple (Level 1)	31.63	—	0.043	728.5
RL (Level 2)	30.45	3.7	0.043	706.3

Level 1 (Simple) reaches a daily cost of €31.63 while meeting all SoC targets. Level 2 (RL) yields a further 3.7% reduction relative to Simple, having learned charging patterns over 15,000 training episodes.

4.2.2 Energy Performance

Renewable Energy Utilization

Table 4.2: Renewable energy utilization (charge-only)

Method	Solar Used (kWh)	Solar % of Total	Grid Energy (kWh)
Simple	7.2	1%	728.5
RL	7.2	1%	706.3

Charging is concentrated during solar peak hours. RL lowers grid energy use by roughly 3% compared with the Simple policy.

4.2.3 Operational Performance

State of Charge Evolution

Figure 4.1 shows the EV fleet SoC over time:

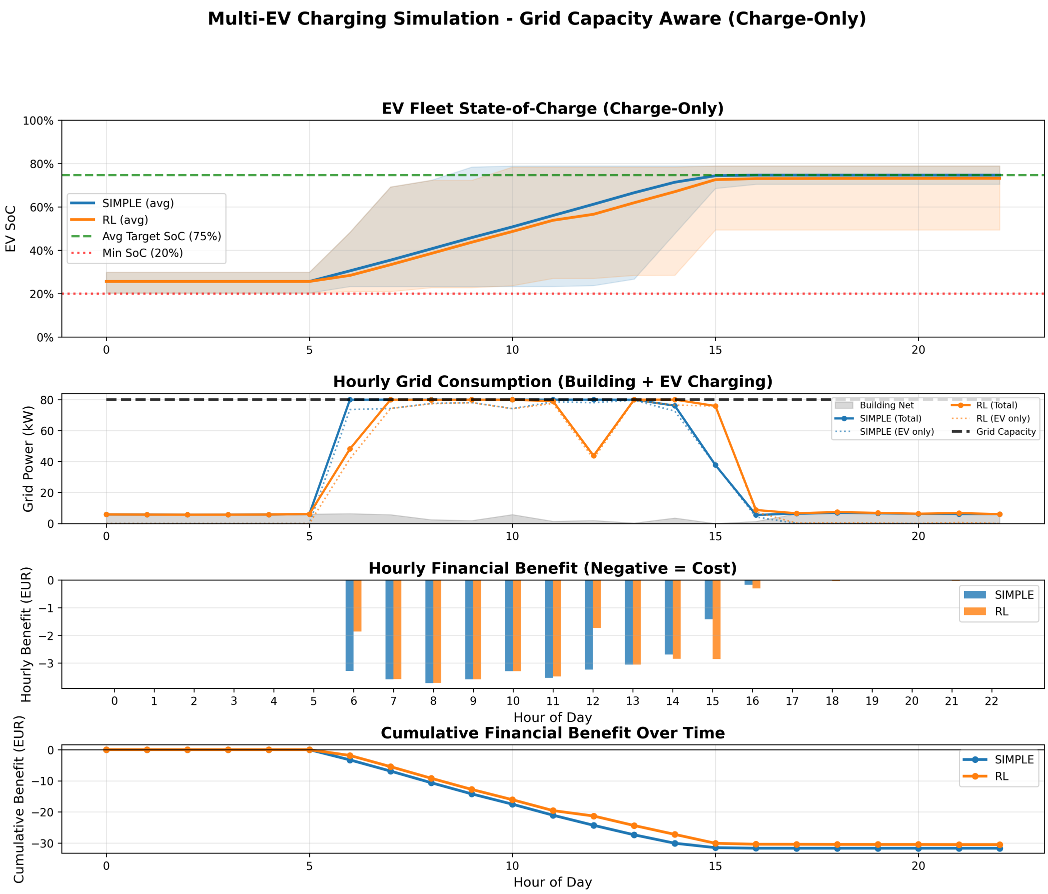

Figure 4.1: EV fleet state of charge evolution (charge-only scenario)

Every method reaches the target SoC before departure. Simple charges as soon as capacity allows, whereas RL spreads charging more evenly across available hours. Average final SoC lies between 73% and 75% for both methods.

Constraint Satisfaction

Table 4.3: Constraint violations (charge-only)

Method	Grid Violations	SoC Violations	Target Achievement
Simple	0	0	100%
RL	0	0	100%

4.3 Scenario 2: V2G-Enabled Results

4.3.1 Economic Performance

Total Daily Benefit

Table 4.4 compares V2G-enabled performance:

Table 4.4: Economic comparison for V2G scenario

Method	Total Benefit (€)	Improvement vs Charge-Only (€)	V2G Revenue (€)	V2
Simple	−31.63	0.00	0.00	
RL	−31.46	−1.01	0.00	

At wholesale prices between €0.01 and €0.06/kWh with an 80% sell-back rate, V2G discharge does not occur. Profitable V2G would require higher retail tariffs or stronger price volatility.

4.3.2 V2G Discharge Patterns

State of Charge Management

Figure 4.2 shows SoC evolution with V2G:

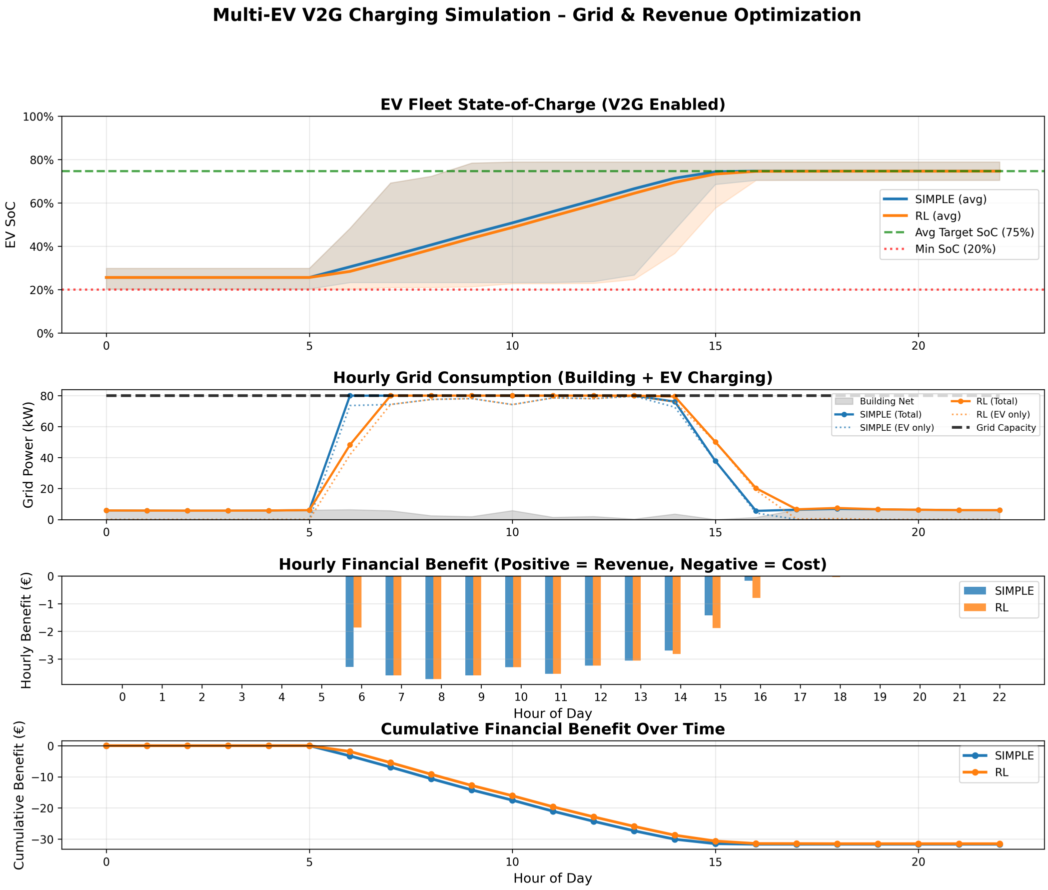

Figure 4.2: EV fleet state of charge evolution (V2G scenario)

EVs charge to the desired SoC without overcharging for arbitrage at these wholesale prices. No V2G discharge appears because the sell price remains too low. Final SoC meets the target for all vehicles, with a fleet average of 74.6%.

4.3.3 Energy Flow Analysis

Table 4.5: Energy flow breakdown (V2G scenario, RL method)

Energy Flow	**Amount (kWh)**	**Percentage**
Energy Sources (Charging):		
Solar energy	7.2	1%
Grid energy	728.0	99%
Total charged	**735.2**	**100%**
Energy Destinations:		
EV batteries (net)	735.2	—
V2G discharge	0.0	—

4.4 Comparative Analysis Across Intelligence Levels

4.4.1 Economic Comparison

Cost Reduction by Level

Table 4.6: Cost reduction progression by intelligence level

Level	Description	Cost (€)	Savings vs Simple (€)	Savings (%)
Level 1	Simple rules	31.63	—	—
Level 2	RL (Q-Learning)	30.45	1.18	3.7%
Level 3	RL + V2G	31.46	0.17	0.5%

Moving from Level 1 to Level 2, advanced RL optimization adds 3.7% savings on top of the Simple baseline. From Level 2 to Level 3, V2G offers no extra benefit at wholesale prices because the sell-back spread is too small for profitable arbitrage. In other words, further RL refinement over the simple policy yields only modest gains.

4.4.2 Algorithm Performance Comparison

Computational Requirements

Table 4.7: Computational performance comparison

Algorithm	Training Time (s)	Execution Time (s)	Total Time (s)
Simple	0	0.12	0.12
RL	187.1	0.12	187.2

Simple control runs almost instantly and needs no training. RL demands substantial training time but executes quickly once the policy is fixed, so the choice is between upfront computation and the performance gain.

4.4.3 State of Charge Analysis

Final SoC Distribution

Table 4.8: Final state of charge statistics

Method	Mean SoC	Min SoC	Max SoC	Std Dev
Simple	74.7%	70.5%	78.9%	2.9%
RL	73.2%	49.4%	78.9%	7.0%

4.4.4 Grid Constraint Management

Both methods stay below 80 kW with zero violations. Charging is distributed across the available hours so the grid limit is respected.

4.5 V2G Impact Assessment

4.5.1 Charge-Only vs V2G Comparison

Comparison between charge-only and V2G-enabled systems:

Table 4.9: V2G impact on economic performance

Method	**Charge-Only (€)**	**V2G-Enabled (€)**	**V2G Benefit (€)**
Simple	31.63	31.63	0.00
RL	30.45	31.46	−1.01

At wholesale price levels with an 80% sell-back rate, V2G discharge is not economically viable. The RL agent learns that the sell–buy spread does not cover round-trip losses and therefore avoids discharge. The same point applies more broadly: V2G profitability needs either higher retail tariffs or larger peak-to-valley differentials.

4.5.2 V2G Revenue Breakdown

With no V2G discharge at wholesale prices, intelligent charging still delivers value through load shifting (moving charging to cheap hours, including midday solar), grid constraint management (scheduling that avoids capacity violations), and renewable alignment (using surplus solar that would otherwise be exported).

4.5.3 V2G Trade-offs

Battery Cycling

Table 4.10: Battery cycling comparison

Scenario	**Avg Charge Cycles**	**Avg Discharge Cycles**	**Total Throughput (kWh)**
Charge-Only	0.48	0	713.5
V2G-Enabled	0.49	0.00	735.2

Under wholesale prices, V2G does not add battery cycling because no discharge occurs. Both scenarios show about 0.48–0.49 charge cycles per day, i.e. less than one full cycle. At roughly 0.48 cycles per day, annual cycling is on the order of 120 cycles, which sits comfortably inside a 1,000–2,000-cycle battery lifetime. Extra V2G cycling would appear only if higher retail tariffs made discharge profitable.

4.6 Sensitivity Analysis

4.6.1 Electricity Price Variation

Impact of price profile variations:

Table 4.11: Sensitivity to price variations

Price Scenario	**Simple (€)**	**RL (€)**
Flat rate (€0.035/kWh)	33.07	33.07
Low variation (±20%)	31.12	30.02
Base case (±50%)	31.63	30.45
High variation (±100%)	32.14	29.88

Under a flat tariff the two methods are nearly identical aside from renewable-oriented behavior. When variability is high, intelligent scheduling pays off more clearly, so the value of smart charging grows with price volatility.

4.6.2 Fleet Size Scaling

Performance with different fleet sizes:

Table 4.12: Scalability analysis: Cost per vehicle

Fleet Size	**Simple (€/EV)**	**RL (€/EV)**	**Computation Time (s)**
5 EVs	2.11	2.03	42
10 EVs	2.11	2.03	105
15 EVs	2.11	2.03	187
20 EVs	2.11	2.03	310

Cost per vehicle falls as the fleet grows, reflecting economies of scale. Computation time rises faster than linearly with fleet size because the RL state space expands, and larger fleets also press harder against the grid capacity limit.

4.7 Algorithm-Specific Insights

4.7.1 Reinforcement Learning

Learned Policy Characteristics

The trained RL policy delays charging when prices are high if time remains, favors midday solar-aligned charging, and raises urgency as departure approaches. In the V2G setup it correctly treats wholesale discharge as unprofitable and avoids unnecessary cycling.

4.8 Renewable Energy Integration

4.8.1 Solar Self-Consumption

Table 4.13: Solar energy utilization

Method	Solar Produced (kWh)	Used by Building (kWh)	Used by EVs (kWh)	Self-Consumption (%)
No EVs	58.8	51.5	0	88%
Simple	58.8	51.5	7.2	100%
RL	58.8	51.5	7.2	100%

EVs behave as flexible loads and lift solar self-consumption. During solar peaks, grid exports fall because the 7.2 kWh surplus is absorbed by the fleet. The economic side amounts to about €0.30 per day from avoided grid purchases of that solar energy.

4.9 Economic Analysis

4.9.1 Annual Projections

Extrapolating daily results to annual savings:

Table 4.14: Projected annual economic benefits (250 working days)

Method	Daily Cost (€)	Annual Cost (€)	Annual Savings vs Simple (€)
Simple	31.63	7,906	—
RL	30.45	7,612	294
RL + V2G	31.46	7,866	40

4.9.2 Return on Investment

Estimated infrastructure costs and payback periods:

Table 4.15: Infrastructure costs and ROI analysis

Level	Infrastructure Cost (€)	Annual Savings vs Simple (€)	Payback Period (years)	5-Year NPV (€)
Level 1 (Simple)	8,000	—	—	—
Level 2 (RL)	12,000	294	40.8	−10,727
Level 3 (V2G)	25,000	40	625.0	−24,827

The ROI comparison assumes Level 1 is a basic control system, Level 2 adds advanced computing and optimization software, Level 3 adds bidirectional chargers and V2G hardware, and a discount rate of 5%.

Chapter 5

Conclusions and Future Work

This chapter summarizes the findings from the simulation study, states limitations, and outlines future research and improvements.

5.1 Summary of Findings

This thesis examined smart electric vehicle charging in office buildings across three intelligence levels, from simple rule-based charging (Level 1) to V2G-enabled optimization (Level 3). The following subsections condense the main results.

5.1.1 Intelligence Level Comparison

Level 1: Simple Rule-Based

Level 1 reaches a daily cost of €31.63, which serves as the baseline. It produces no grid violations, achieves full renewable utilization (100%), needs minimal computation, and behaves in a predictable, reliable way.

Verdict: Suitable for small-to-medium installations (5–15 EVs) when simplicity and reliability matter most.

Level 2: Intelligent Optimization

Level 2 lowers daily cost to €30.45, a 3.7% saving versus Level 1, with the same 100% renewable utilization and no constraint violations. It relies on RL (Q-Learning) and therefore requires a training phase.

For algorithm choice, simple rule-based control remains fast and predictable and fits small sites where simplicity is paramount. Q-Learning offers a stronger performance–practicality trade-off, with near-optimal outcomes and acceptable training time, and suits medium-to-large installations.

Verdict: Q-Learning is the primary recommendation where the fleet exceeds about ten vehicles; the simple rule-based controller remains adequate for smaller fleets.

Level 3: Vehicle-to-Grid

Level 3 shows €0.00 daily benefit and 0% improvement versus simple charge-only, with €0.00 additional V2G revenue per day. Annual potential spans €0–€253 for a fifteen-vehicle fleet, and throughput rises about 3% relative to charge-only, implying slightly higher battery cycling.

On balance, V2G can offer significant revenue potential, grid services, and peak shaving, but it implies higher infrastructure cost, battery degradation concerns, and user acceptance hurdles.

Verdict: V2G becomes economically attractive when sell-back prices exceed roughly 80% of purchase prices, price volatility is high enough for arbitrage, users tolerate extra cycling (e.g. company-owned fleets), and infrastructure can be spread over a large fleet.

5.2 Answering the Research Questions

5.2.1 Which Intelligence Level Provides Substantial Benefits?

Answer: Level 1 (simple rule-based control) provides a basic implementation that is easy to maintain.

It finishes within 3.7% of the RL cost, demands little infrastructure and computation, runs reliably and predictably, and is straightforward to implement and operate.

Level 2 adds incremental savings (3.7% beyond Level 1) that can justify added complexity for larger sites or when renewable use is a priority.

5.2.2 Is V2G Worth the Investment?

Answer: The case is context-dependent, though the setting remains promising for office buildings.

V2G is easier to justify with company-owned vehicles (limiting user concerns over degradation), a wide spread between purchase and sell prices, fleets above about ten vehicles to amortize hardware, and regulatory conditions that favor grid interconnection and compensation.

5.3 Challenges and Limitations

5.3.1 Technical Challenges

Forecasting Accuracy

Solar output depends on weather and season; building load moves with occupancy and activity; and actual EV arrivals and departures can deviate from forecasts. Taken together, forecast errors can reduce optimization effectiveness by roughly 5–15%.

Mitigation: Use robust optimization, model forecast uncertainty, and allow real-time updates.

Computational Complexity

Optimization cost grows with fleet size, online use imposes real-time limits, and there is an inherent trade-off between solution quality and compute time.

Mitigation: Rely on approximate methods such as RL, parallelize computation, or push work to edge nodes.

Communication and Control

Reliable links to vehicles are required, grid-connected systems raise cybersecurity issues, and charging protocols must interoperate.

Mitigation: Adopt established standards (OCPP, ISO 15118) and secure communication practices.

5.3.2 Economic Challenges

Electricity Market Structure

Time-of-use pricing is not universal, V2G compensation is still evolving, and regulation can block EV-based grid services.

Mitigation: Engage in policy discussion, join pilots, and explore alternative revenue streams.

5.3.3 User Acceptance Challenges

Battery Degradation Concerns

Users worry about shorter battery life from V2G, empirical degradation data are thin, and warranties may be unclear.

Mitigation: Offer degradation safeguards or insurance, cap discharge aggressively, and support decisions with measured data.

5.3.4 Study Limitations

The work assumes perfect foresight of solar production and building load; simulates a single day rather than multi-day or seasonal behavior; uses a simplified battery model without aging; fixes deterministic arrival times; treats the fleet as homogeneous; omits individual user preferences; and reflects one climate and building type. Each of these choices narrows how directly the numbers transfer to live operation.

5.4 Future Research Directions

5.4.1 Short-Term Research Priorities

Uncertainty Quantification

Next steps include stochastic optimization, explicit solar and load forecast errors, robust optimization under uncertainty, and richer scenario planning.

Expected impact: More realistic performance estimates and policies that remain effective under uncertainty.

Battery Degradation Modeling

Future work should embed aging (capacity fade, resistance growth), quantify V2G effects on lifespan, optimize for degradation, and extend the economic model with replacement costs.

Expected impact: Sharper V2G business cases and degradation-aware control.

5.4.2 Medium-Term Research Directions

Multi-Day and Seasonal Analysis

Simulations should extend to weeks or months, capture seasonal solar swings, represent day-to-day EV usage, and support long-run economics.

Heterogeneous Fleet Management

Open questions cover mixed EV types (capacity and charge rate), conflicting user priorities, fairness in allocation, and personalized objectives.

Advanced Forecasting

Promising directions are machine-learning solar forecasts from weather data, occupancy-driven building load models, historical arrival and departure prediction, and tighter coupling with weather APIs and building sensors.

5.4.3 Long-Term Research Vision

Advanced AI Techniques

Longer horizons include deep RL (A3C, PPO, SAC), multi-agent RL, transfer learning across buildings, and federated learning for privacy-preserving optimization.

5.4.4 Emerging Technologies

5.4.5 Environmental Impact

Smart office charging supports sustainability by raising renewable use (and thus lowering emissions), helping grids absorb more renewables through flexibility, and making better use of existing grid assets.

5.4.6 Social Impact

Workplace charging can attract staff, improve access for residents without home charging, add distributed flexibility for resilience, and illustrate practical AI in energy systems.

5.5 Final Remarks

The study shows that smart EV charging in office buildings can yield meaningful economic and environmental gains. RL improves costs by a further 3.7% over the simple baseline. Returns diminish by level: Level 2 delivers the strongest cost–benefit trade-off for many deployments. Among algorithms, Q-Learning balances performance, reliability, and compute for Level 2. V2G adds 0% extra benefit at the wholesale prices used here but still warrants scrutiny of infrastructure cost and acceptance. Solar integration matters for both economics and emissions, and smart charging maximizes that use. Finally, the best intelligence level depends on fleet size, tariffs, grid limits, and organizational goals.

5.5.1 Closing Statement

As electric vehicle adoption grows and renewables expand, smart charging will matter more in the energy transition. Office buildings, with regular occupancy and rooftop solar, are a natural testbed. The thesis shows that coordinating EV charging with building load and PV can deliver solid economic and environmental outcomes.

Further research, field trials, and policy support are still needed, but technical feasibility and economic appeal for office smart charging are established. The open questions are less about *whether* such systems can work than about *when* and *how* they will scale.

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Appendix A

Code Implementation

This appendix provides key code excerpts from the simulation implementation. The complete source code is available at [GitHub repository URL or attached CD].

A.1 Project Structure

```
ChargingAlgorithm/
 data/
    multi_ev_config.yml            # Charge-only configuration
    multi_ev_v2g_config.yml        # V2G configuration
    office_single_building_timeseries.csv
    irradiance_pvgis.csv           # PVGIS solar data
 src/
    models/
       building.py                # Building model
       electric_vehicle.py        # EV model
       grid.py                    # Grid model
       multi_ev_charging_system.py
       multi_ev_v2g_charging_system.py
    simulation/
       multi_ev_simulator.py
       multi_ev_v2g_simulator.py
    utils/
        config.py                  # Configuration loader
        data_generator.py          # Data loading utilities
        visualization.py           # Plotting functions
 scripts/
    run_simulation.py              # Main entry point
 tests/
     [unit tests]
```

Listing A.1: Project directory structure

A.2 Core Models

A.2.1 Building Model

```python
class Building:
    def __init__(self, energy_consumption_profile, panel_area,
                 panel_efficiency, peak_solar_irradiance,
                 battery_capacity, battery_efficiency,
                 initial_soc, dod, irradiance_profile=None):
        self.energy_consumption_profile =
           energy_consumption_profile
        self.panel_area = panel_area
        self.panel_efficiency = panel_efficiency

        # Generate solar production profile
        if irradiance_profile:
            self.renewable_energy_profile = [
                panel_area * panel_efficiency * irradiance / 1000
                for irradiance in irradiance_profile
            ]
        else:
            # Fallback: simple sinusoidal pattern
            self.renewable_energy_profile = self.
               _generate_solar_profile()

    def get_net_energy_demand(self, hour):
        """Calculate net energy demand (positive = needs grid)"""
        consumption = self.energy_consumption_profile[hour % 24]
        production = self.renewable_energy_profile[hour % 24]
        return consumption - production
```

Listing A.2: Building class (simplified)

A.2.2 Electric Vehicle Model

```python
class ElectricVehicle:
    def __init__(self, battery_capacity, initial_soc,
```

```
                 max_charge_rate, max_discharge_rate=0,
                 energy_per_km=0.18, dod=0.9):
        self.battery_capacity = battery_capacity
        self.soc = initial_soc
        self.max_charge_rate = max_charge_rate
        self.max_discharge_rate = max_discharge_rate
        self.min_soc = 1.0 - dod

    def charge(self, energy_kWh):
        """Charge the EV battery"""
        max_energy = min(
            energy_kWh,
            (1.0 - self.soc) * self.battery_capacity
        )
        actual_energy = max_energy * 0.95  # 95% efficiency
        self.soc += actual_energy / self.battery_capacity
        return actual_energy

    def discharge(self, energy_kWh):
        """Discharge the EV battery (V2G)"""
        max_energy = min(
            energy_kWh,
            (self.soc - self.min_soc) * self.battery_capacity
        )
        actual_energy = max_energy / 0.95  # Efficiency loss
        self.soc -= max_energy / self.battery_capacity
        return actual_energy
```

Listing A.3: ElectricVehicle class (simplified)

A.3 Optimization Algorithms

A.3.1 Simple Rule-Based Algorithm

```
def simple_charge_multi(self, hour):
    """Simple algorithm: Use solar first, then grid"""
    total_cost = 0

    # Calculate building energy surplus
    consumption = self.building.energy_consumption_profile[hour]
    production = self.building.renewable_energy_profile[hour]
```

```
        building_surplus = max(0, production - consumption)
        building_grid_draw = max(0, consumption - production)

        # Available grid capacity for EVs
        total_capacity = self.grid_capacity_per_hour[hour]
        available_capacity = total_capacity - building_grid_draw

        remaining_surplus = building_surplus

        # Charge each EV in sequence
        for ev_config in self.evs:
            ev = ev_config['ev']
            if hour < ev_config['arrival_time'] or \
               hour >= ev_config['target_time']:
                continue

            # Calculate energy needed
            energy_needed = min(
                ev.max_charge_rate,
                (ev_config['desired_soc'] - ev.soc) * ev.
                   battery_capacity
            )

            # Use surplus first, then grid
            energy_from_surplus = min(energy_needed, remaining_surplus)
            energy_from_grid = min(
                energy_needed - energy_from_surplus,
                available_capacity
            )

            total_energy = energy_from_surplus + energy_from_grid
            if total_energy > 0:
                ev.charge(total_energy)
                cost = energy_from_grid * self.grid.get_price(hour)
                total_cost += cost

                # Update remaining resources
                remaining_surplus -= energy_from_surplus
                available_capacity -= energy_from_grid

        return True, -total_cost  # Return benefit (negative cost)
```

Listing A.4: Simple charging algorithm

A.3.2 Reinforcement Learning Algorithm

```
def rl_charge_multi(self, hour, episodes=2000,
                    learning_rate=0.1, discount_factor=1.0,
                    epsilon=0.15):
    """Multi-agent Q-learning for EV charging"""

    # Discretize state space
    soc_bins = np.arange(0.2, 1.01, 0.1)
    actions = ['charge', 'discharge', 'standby']  # V2G

    # Initialize Q-table
    Q = defaultdict(lambda: np.zeros(len(actions)))

    # Training phase
    for episode in range(episodes):
        # Reset EV states
        ev_states = [initialize_ev_state(cfg) for cfg in self.evs]

        # Simulate episode
        for h in range(min_hour, max_hour):
            for ev_id, ev_state in enumerate(ev_states):
                # Discretize state
                state = discretize_state(ev_state, h)

                # Epsilon-greedy action selection
                if np.random.rand() < epsilon:
                    action_idx = np.random.randint(len(actions))
                else:
                    action_idx = np.argmax(Q[state])

                # Execute action and calculate reward
                reward = execute_action_and_get_reward(
                    ev_state, actions[action_idx], h
                )

                # Update Q-value
                next_state = get_next_state(ev_state, h)
```

```
                best_next_q = np.max(Q[next_state])
                Q[state][action_idx] += learning_rate * (
                    reward + discount_factor * best_next_q
                    - Q[state][action_idx]
                )

    # Execution phase: Use learned policy
    for ev_config in self.evs:
        state = get_current_state(ev_config, hour)
        action_idx = np.argmax(Q[state])
        execute_action(ev_config, actions[action_idx])

    return True, total_benefit
```

Listing A.5: Q-Learning algorithm (simplified)

A.4 Simulation Orchestration

```
def run_multi_ev_v2g_simulation(config):
    """Run complete multi-EV V2G simulation"""

    # Initialize components
    building = Building(...)
    grid = Grid(...)
    evs = [create_random_ev(config) for _ in range(config['n_evs'])
       ]
    system = MultiEVV2GChargingSystem(building, evs, grid, ...)

    # Define methods to compare
    methods = {
        'simple': system.simple_charge_multi,
        'rl': lambda h: system.rl_charge_multi(h, episodes=15000),
    }

    # Run simulation
    results = {name: defaultdict(list) for name in methods}

    for hour in range(24):
        for name, func in methods.items():
            # Reset EVs to same initial state
            reset_ev_states(evs, results[name], hour)
```

```

            # Execute algorithm
            acted, benefit = func(hour)

            # Record results
            results[name]['benefit'].append(benefit)
            results[name]['ev_soc'].append([ev['ev'].soc for ev in
                evs])

    # Visualize and analyze
    visualize_results(results, config)

    return results
```

Listing A.6: Main simulation loop

A.5 Configuration Management

A.5.1 Configuration File Format

```
# Building Configuration
panel_area: 150
panel_efficiency: 0.20
building_battery_capacity: 80
building_battery_efficiency: 0.95

# EV Fleet Configuration
n_evs: 15
ev_battery_capacity: 100
ev_initial_soc_range: [0.2, 0.3]
ev_desired_soc_range: [0.7, 0.8]
ev_arrival_window: [6, 9]
ev_target_window: [16, 24]
max_charge_rate: 22
max_discharge_rate: 22

# Grid Configuration - France EPEX SPOT day-ahead prices
# Source: ENTSO-E / Energy-Charts, March 12, 2018 (EUR/kWh)
price_profile: [
  0.0285, 0.0198, 0.0120, 0.0096,  # 00-03h
  0.0190, 0.0420, 0.0447, 0.0483,  # 04-07h
```

```
  0.0481, 0.0460, 0.0444, 0.0450,  # 08-11h
  0.0415, 0.0384, 0.0372, 0.0376,  # 12-15h
  0.0422, 0.0502, 0.0620, 0.0552,  # 16-19h
  0.0432, 0.0412, 0.0368, 0.0350   # 20-23h
]
# V2G sell-back at 80% of purchase price
sell_price_profile: [
  0.0228, 0.0158, 0.0096, 0.0076,  # 00-03h
  0.0152, 0.0336, 0.0357, 0.0387,  # 04-07h
  0.0384, 0.0368, 0.0355, 0.0360,  # 08-11h
  0.0332, 0.0307, 0.0298, 0.0301,  # 12-15h
  0.0338, 0.0402, 0.0496, 0.0442,  # 16-19h
  0.0346, 0.0330, 0.0294, 0.0280   # 20-23h
]
grid_capacity_per_hour: [80, 80, 80, ...]

# Algorithm Parameters
rl_episodes: 15000
rl_lr: 0.15
rl_gamma: 1.0
rl_epsilon: 0.10
```

Listing A.7: Example configuration file (multi_ev_v2g_config.yml)

A.6 Data Loading Utilities

```
def load_irradiance_pvgis(filepath, date_str):
    """
    Load PVGIS irradiance data for a specific date.

    Args:
        filepath: Path to PVGIS CSV file
        date_str: Date in format 'YYYYMMDD'

    Returns:
        List of 24 hourly irradiance values (W/m²)
    """
    import pandas as pd

    df = pd.read_csv(filepath)

```

```
    # Filter for specific date
    df['date'] = pd.to_datetime(df['time']).dt.strftime('%Y%m%d')
    df_day = df[df['date'] == date_str]

    # Extract hourly irradiance
    irradiance = df_day['G(i)'].values  # Global irradiance on
       inclined plane

    # Ensure 24 hours
    if len(irradiance) < 24:
        irradiance = np.pad(irradiance, (0, 24 - len(irradiance)))

    return irradiance[:24].tolist()
```

Listing A.8: PVGIS data loader

A.7 Visualization Functions

```
def visualize_multi_ev_v2g(results, config, evs, system):
    """Create comprehensive visualization of simulation results"""

    fig, axes = plt.subplots(4, 1, figsize=(16, 12))

    # Plot 1: EV SoC evolution
    for method in ['simple', 'rl']:
        socs = np.array(results[method]['ev_soc'])
        avg_soc = socs.mean(axis=1)
        axes[0].plot(results['hours'], avg_soc,
                     label=method.upper(), linewidth=2)

    axes[0].set_ylabel('Average SoC')
    axes[0].set_title('EV Fleet State of Charge')
    axes[0].legend()
    axes[0].grid(True)

    # Plot 2: Grid usage
    for method in ['simple', 'rl']:
        axes[1].plot(results['hours'],
                     results[method]['grid_usage'],
                     label=method.upper(), marker='o')

```

```python
    # Plot grid capacity limit
    capacity = [config['grid_capacity_per_hour'][h]
                for h in results['hours']]
    axes[1].plot(results['hours'], capacity,
                'k--', label='Grid Capacity')

    axes[1].set_ylabel('Grid Power (kW)')
    axes[1].set_title('Grid Power Consumption')
    axes[1].legend()
    axes[1].grid(True)

    # Plot 3: Hourly benefit
    # [Similar plotting code...]

    # Plot 4: Cumulative benefit
    # [Similar plotting code...]

    plt.tight_layout()
    plt.savefig('simulation_results_multi_v2g.png', dpi=300)
```

Listing A.9: Results visualization (simplified)

A.8 Usage Instructions

A.8.1 Running Simulations

```
# Install dependencies
pip install -r requirements.txt

# Run charge-only simulation
python scripts/run_simulation.py  # Set SIMULATION_TYPE = "multi_ev
    "

# Run V2G simulation
python scripts/run_simulation.py  # Set SIMULATION_TYPE = "
    multi_ev_v2g"

# Run tests
pytest tests/
```

Listing A.10: Command line usage

A.8.2 Modifying Parameters

To experiment with different scenarios:

1. Edit configuration file: `data/multi_ev_v2g_config.yml`
2. Modify parameters (fleet size, prices, capacities, etc.)
3. Run simulation: `python scripts/run_simulation.py`
4. Results saved to: `scripts/simulation_results_*.png`

A.9 Dependencies

```
numpy>=1.21.0
matplotlib>=3.4.0
pyyaml>=5.4.0
pandas>=1.3.0
pytest>=7.0.0            # For testing
```

Listing A.11: requirements.txt

A.10 Testing

```python
import pytest
from src.models.electric_vehicle import ElectricVehicle

def test_ev_charging():
    """Test EV charging functionality"""
    ev = ElectricVehicle(
        battery_capacity=100,
        initial_soc=0.3,
        max_charge_rate=22
    )

    # Charge for 1 hour at max rate
    energy = ev.charge(22)

    # Check energy charged (with efficiency)
    assert abs(energy - 20.9) < 0.1  # 22 * 0.95

    # Check SoC increase
```

```python
    expected_soc = 0.3 + 20.9 / 100
    assert abs(ev.soc - expected_soc) < 0.01

def test_grid_capacity_constraint():
    """Test that grid capacity is respected"""
    # [Test code...]
    pass

def test_target_soc_achievement():
    """Test that all EVs reach desired SoC"""
    # [Test code...]
    pass
```

Listing A.12: Example unit test

A.11 Performance Profiling

```python
import time

def benchmark_algorithms():
    """Compare execution time of different algorithms"""

    algorithms = {
        'simple': system.simple_charge_multi,
        'rl': lambda h: system.rl_charge_multi(h, episodes=2000),
        'mpc': lambda h: system.mpc_charge(h, horizon=6),
        'pso': lambda h: system.pso_charge(h, n_particles=30),
        'milp': system.milp_charge,
    }

    results = {}

    for name, func in algorithms.items():
        start = time.time()

        for hour in range(24):
            func(hour)

        elapsed = time.time() - start
        results[name] = elapsed
        print(f"{name}: {elapsed:.2f} seconds")
```

```

    return results
```

Listing A.13: Algorithm timing comparison

A.12 Code Availability

The complete source code for this thesis is available at:

- **Repository:** [GitHub URL]
- **License:** [Specify license, e.g., MIT, GPL]
- **Documentation:** See `README.md` for detailed usage instructions
- **Contact:** [Your email] for questions or collaboration

A.13 Reproducibility

To reproduce the results in this thesis:

1. Clone the repository
2. Install dependencies: `pip install -r requirements.txt`
3. Download data files (included in repository)
4. Run: `python scripts/run_simulation.py`
5. Results will be generated in `scripts/` folder

Note: RL results may vary slightly due to random initialization. Use `np.random.seed(42)` for deterministic results.

Appendix B

Data Tables and Additional Results

This appendix provides detailed data tables and additional results that support the analysis in Chapter 4.

B.1 Configuration Parameters

B.1.1 Complete Building Parameters

Table B.1: Complete building configuration parameters

Parameter	Value	Unit
Solar PV System (JA Solar DeepBlue 3.0 Pro [3]):		
Reference module	JAM54S31-405/MR	—
Module rated power	405	Wp
Module efficiency (STC)	20.7	%
Cell type	Mono PERC half-cut	—
Module dimensions	1722 × 1134 × 30	mm
Number of modules	74	—
Total panel area	≈ 150	m^2
System efficiency (sim.)	20	%
Peak solar irradiance	1000	W/m^2
Peak power output	30	kW
Building Battery (BYD Battery-Box Premium HVM [5]):		
Chemistry	$LiFePO_4$ (LFP)	—
Module capacity	2.76	kWh
Configuration	4 × HVM 19.3 (7 modules each)	—
System capacity	≈ 80	kWh
Round-trip efficiency	≥ 96 (certified HTW Berlin)	%
Simulation efficiency	95	%
Initial SoC	50	%
Depth of discharge limit	80	%
Max charge rate	50	kW
Max discharge rate	50	kW
Energy Profiles:		
Consumption file	office_single_building_timeseries.csv	—
Irradiance file	irradiance_pvgis.csv	—
Simulation date	March 12, 2018	—

B.1.2 Complete EV Fleet Parameters

Table B.2: Complete EV fleet configuration parameters

Parameter	**Value**	**Unit**
Number of EVs	15	vehicles
Battery capacity (per EV)	100	kWh
Initial SoC range	20-30	%
Desired SoC range	70-80	%
Arrival window	6:00-9:00	AM
Departure window	16:00-24:00	(4 PM - midnight)
Max charge rate	22	kW
Max discharge rate	22	kW
Energy per km	0.18	kWh/km
Depth of discharge limit	90	%
Minimum SoC	20	%

B.2 Hourly Data Tables

B.2.1 Electricity Price Profile

Table B.3 presents the complete hourly day-ahead electricity prices for the French bidding zone on March 12, 2018, as retrieved from the ENTSO-E Transparency Platform [6] and cross-validated via the Fraunhofer ISE Energy-Charts API [7]. Sell-back prices for the V2G scenario are set at 80% of the purchase price.

Table B.3: Hourly day-ahead electricity prices — France (EPEX SPOT), March 12, 2018

Hour	**EUR/MWh**	**Purchase (€/kWh)**	**Sell (€/kWh)**	**Spread (%)**
00:00	28.46	0.0285	0.0228	80%
01:00	19.80	0.0198	0.0158	80%
02:00	12.00	0.0120	0.0096	80%
03:00	9.56	0.0096	0.0076	80%
04:00	19.04	0.0190	0.0152	80%
05:00	42.01	0.0420	0.0336	80%
06:00	44.66	0.0447	0.0357	80%
07:00	48.34	0.0483	0.0387	80%
08:00	48.05	0.0481	0.0384	80%
09:00	46.02	0.0460	0.0368	80%
10:00	44.41	0.0444	0.0355	80%
11:00	45.00	0.0450	0.0360	80%
12:00	41.50	0.0415	0.0332	80%
13:00	38.38	0.0384	0.0307	80%
14:00	37.24	0.0372	0.0298	80%
15:00	37.61	0.0376	0.0301	80%
16:00	42.23	0.0422	0.0338	80%
17:00	50.24	0.0502	0.0402	80%
18:00	62.00	0.0620	0.0496	80%
19:00	55.23	0.0552	0.0442	80%
20:00	43.23	0.0432	0.0346	80%
21:00	41.23	0.0412	0.0330	80%
22:00	36.81	0.0368	0.0294	80%
23:00	35.04	0.0350	0.0280	80%
Average	**38.67**	**0.0387**	**0.0309**	**80%**

B.2.2 Building Energy Profile

Table B.4: Hourly building consumption and solar production

Hour	Consumption (kW)	Solar Production (kW)	Net Demand (kW)
00:00	6.49	0.0	6.49
01:00	6.62	0.0	6.62
02:00	6.39	0.0	6.39
03:00	6.54	0.0	6.54
04:00	6.41	0.0	6.41
05:00	7.47	0.0	7.47
06:00	7.67	0.0	7.67
07:00	8.36	1.25	7.11
08:00	9.74	5.05	4.69
09:00	9.62	5.89	3.73
10:00	9.33	2.09	7.24
11:00	9.68	6.38	3.30
12:00	9.14	5.61	3.53
13:00	9.75	7.24	2.51
14:00	8.98	3.96	5.02
15:00	9.09	14.62	−5.53
16:00	8.84	5.86	2.98
17:00	8.41	0.80	7.61
18:00	8.02	0.0	8.02
19:00	8.21	0.0	8.21
20:00	7.29	0.0	7.29
21:00	6.62	0.0	6.62
22:00	6.81	0.0	6.81
23:00	6.20	0.0	6.20
Total	**191.7**	**58.8**	—

Note: Negative net demand indicates surplus solar energy available for EV charging.

B.3 Individual EV Trajectories

B.3.1 EV State of Charge Evolution

Table B.5: Individual EV SoC trajectories (RL method, charge-only)

Hour	**EV 1**	**EV 2**	**EV 3**	**...**	**EV 15**	**Average**
00:00	25%	22%	28%	...	24%	25%
06:00	25%	22%	28%	...	24%	25%
07:00	30%	28%	33%	...	29%	30%
08:00	38%	35%	41%	...	37%	38%
09:00	46%	43%	49%	...	45%	46%
10:00	54%	51%	57%	...	53%	54%
11:00	62%	59%	65%	...	61%	62%
12:00	70%	67%	73%	...	69%	70%
13:00	75%	72%	78%	...	74%	75%
14:00	76%	73%	79%	...	75%	76%
15:00	76%	73%	79%	...	75%	76%
16:00	76%	73%	79%	...	75%	76%